Theoretical Formulation & Physics Derivations

Commercial Sub-Critical Hybrid Fission-Fusion Reactor Paradigm

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1. 1D 3-Temperature Hydrodynamics & Thermonuclear Fusion

The staggered-grid 1D radial Lagrangian hydrodynamics conservation laws govern mass, momentum, and ion-electron-radiation thermal energy transport:

Continuity: &partial;ρ/&partial;t + $\nabla \cdot (\rho \mathbf{u}) = 0$
Momentum: $\rho(D\mathbf{u}/Dt) = -\nabla(P_i + P_e + P_r + q_{visc})$
Bosch-Hale D-T Reactivity: ■σv■ = $C_1 \theta \sqrt{\xi / (m_r c^2 T^3)}) \exp(-3\xi)$

2. Sub-Critical Blanket Neutronics & Energy Multiplication

External 14.1 MeV D-T fusion neutrons trigger secondary fission cascades in the sub-critical blanket (k_{eff} = 0.965). The total fissions per source neutron f_{fiss} and blanket energy multiplication $M_{blanket}$ are derived as:

Fission Density: $f_{fiss} = [k_{eff} / (\nu_{bar} (1 - k_{eff}))] \times (1 - f_{stream})$
Energy Multiplier: $M_{blanket} = 1 + f_{fiss} \times (E_{fission} / E_{fusion}) = 1 + f_{fiss} \times (200 \text{ MeV} / 17.6 \text{ MeV})$
Plant Gain: $Q_{electric} = P_{gross} / P_{recirc} = (Q_{fusion} \times M_{blanket} \times \eta_{thermal}) / (\eta_{driver}$■$^{1})$

3. FLiBe Thermal-Hydraulics & First-Wall Mechanics

The lumped coolant heat balance and SDOF spherical shell dynamic hoop stress response equation:

Coolant Balance: $V_{loop} \rho C_p (dT_{out}/dt) = P_{thermal}(t) - \dot{m} C_p (T_{out} - T_{in})$
Hoop Stress SDOF: $d^2\sigma/dt^2 + 2 \zeta \omega_n (d\sigma/dt) + \omega_n^2 \sigma = (P(t) R_{wall} / 2 t_{wall}) \omega_n^2$